

Semiotics and Semantics (Lecture) ~ Lorena Sganzerla ~ Active Inference for the Social Sciences 2023

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everything all right hello and welcome everyone this is the course active inference for the social sciences in 2023. it's August 30th or maybe a different date depending on where you are and today we're in the semiotics and semantics section we'll be hearing a lecture from Lorena's or ziganzela and then we're gonna have a little bit of a discussion so Lorena thank you a lot for joining and off to you for the lecture okay hi everyone um I'm based in Australia yeah I'm Lorena I'm doing actually my PhD in philosophy here I'm working in cognitive sciences and philosophy I work within from Concepts or Innovations of information so I'm very interested in um foundational understanding of cognitive sciences and information and how this this kind of things cash out and this is why I was interested in talking about semantics and social cognition to see how these things cash out and have a few ideas on how semiotics can be compatible with like active imprints and I try to organize a little bit of this semiolic notion and active inference to see if they pan out maybe they don't so let's see how he goes um so let's talk about active inference formal semantics and social cognition thank you what's happening okay I cannot move my slides now we can so I will first have a very short introduction for you guys I have out here so what you're going to do like we have a short introduction I'm going to talk a little bit of theories of meaning and then we're gonna talk a little bit the free energy principle and active inference uh and how shared symbols and social cognition can be thought together I bring one case that I find interesting and then we draw some conclusions so let me know what you think in the comments and after we can have a little discussion or in the next session the background introduction that I want to first point to is we are talking about biological agents and how they act so we're always thinking in terms of biological agents acting to express uh you know that Express informational sensitivities to context dependent relevance to their environment and things uh that involve communities however how this very

minimal way of talking about what motivates and what is relevant to people becomes stable enough as a self-sustaining pattern uh of interactions like are they stable and how they become that stable soon how can those information sensitivities ground values norms and goals like consistently and over time assuming that those values and goals are necessary for Meaningful interactions so we have two line motivations to behavior right that's not really obvious and ultimately how meaning is brought about I don't think I'm gonna answer that but I want to point to some directions and how people have been thinking about these questions over time uh so the problem of meaning has a brief account right like traditionally approaches of uh to cognition localized meaning inside individual organisms and more specifically inside their minds so meaning pertains to content and these kind of things so many have however like many have a developmental history so it's hard to catch those two things together if it's inside people's minds how it can have an evolutionary in developmental history assuming that means are same things that we can learn over time and they happen in shared contexts they don't happen out of nowhere in a vision from nowhere in a sense like we can say that it's not really trivial that conventionalized words like for example mini the conventionalized meaning of a word for instance cat means catch have evolutionary history one thing does not come from the other we need a better understanding to catch this meaning out and that's where I'm stepping from like that's my grounding work you need to catch these things out in terms of community of language users which brings a problem right we have goals and values that can be defined in terms of those communities and you have material and body factors so how these two things are together or how how goals and values affect and motivate material anybody's agents to answer this kind of question like over time there are a few preferred ways like some I would do a brief overview of some theories of meaning they usually in terms of cognitive sciences and any philosophy of mind that's what I'm interested in how they catch it out they usually try to think of those things in terms of syntax like concerns with form as an abstract organizing kind of role it's a type of grammar and how uh at the order of things and how The Logical unfolding of those symbols how they become meaningful and we but the syntax itself is not really enough to cash out many in semantics so then in that syntax evolve into a field that we call semantics in which meaning and mental constant are organized throughout this instructional syntax then we have prepositional knowledge and Truth conditions in which allows these organizational story of meaningful words to make sense and they're going to point out into the into the outside world and we have a third option it's an it's not so new isn't it well it's not so famous which is semantics or more importantly I will focus on biosemantics uh

they use it concerns with the use of symbols like how they bring about meaning in terms of pragmatics and that would be a little bit different and we'll catch it out in different terms in relation to its peers like semantics and syntax well syntax is pretty clear in terms of being rules right so what is semantics talk about so talk about mental content about intentionality about mental representations in the sense we have semantics that have a specific key Concepts that reference truth conditions and denotation so if you have mental content and if you put these meanings inside the organism it has to refer to something else and how this cognitive penetrate penetration happens like how can meaning come about Inside the Mind and how that refers to something outside in the world and how this observational Viewpoint of you thinking about something in front of you can be actually true so we need we need true conditions that denote this specific relationship and that is represented like in terms of mental content contemporarily there's like very ways of catching this out like uh it can that can be functional that can be more an internalized mental specific type of representations grounded in Vegas it's a very rich discussion and I don't want to go into all that but just want to point it out that science and Neuroscience use those terms to study and to build models that relates to mental representations in terms of newer representations because it kind of makes sense for us like brain activity you can see them in terms of what happens by studying human beings and babysitting animals and how these make the representations relate to something that happens in the world you can see that one Behavior elicits specific activities and they have a correlational uh relationship and that is highly debated but there we go we have many types of Cashing Out newer representations and concerning to this relationship of stimuli and the interpretation of the stimuli and how they motivate action that's already a very much weaker claim in relation to mental content and things being inside organisms but it is still in the same line of reasoning and there is also biosity semiotics which I want to focus a little bit more because it concerned with the use and interpretation of signs and symbols which then also emphasizes like the nature the embodied nature and the situated nature of meaning so you don't have meaning without those practices and without the use of embodied agents so we have to reject the idea that meaning are abstract symbols or they kind of be something that has to be in your mind and refer to something outside they're gonna only exist in this type of relationship and they can take many forms including chemicals signals like visual Clues Behavior words like the word catch you can have and how you organized it would get they're all symbolic and that fits also in that kind of structure so they are very important for communication obviously because we do communicate in terms of pointing to the same shared

Notions of symbols in words and also other types of symbolic meaning and we coordinate life around them so living systems are then here characterized by the semiotics but the ability of creating and interpreting and using these symbols in terms of their practices the fathers like of biosemiotics it started like in but probably in the 19th century and Ox coal is one of the most famous Representatives currently we can find new authors like Hoffman or bigger so I try to organize a little bit in a specific model uh in a type of model individual Story how this uh two types of conditioning meaning are being applied in terms of series of meaning historically so even by semiotics we have into pragmatics and you have to practice and they relate to Dynamics and context dependent so you can see here on the top of the screen how agent we have the agent relationship with its psychological and social aspects and by semiotics is somewhat concerned how correlation leads to prediction and beliefs and that leads to expectations and we also have other types of theories likewise uh tele semantics and um and theories of computational syntax that tries to understand the Dynamics and the mechanism of these uh relationship and they are not really Crossing each other they are not in the same they're not talking about the same necessary things so how can these things be brought together that's when we I was thinking how can bring this free energy principle and active inference in terms of those relationships so well different principle and active active influence as we know and I have been talking about probably in the past two classes they have it's a model that is interesting for that because it has like very minimal assumptions so it starts off from the idea that organisms act to maintain themselves in their expected biological and cognitive States so you're always talking you're you're modeling and you're using this mathematical to ground uh biological agents in very embodied sense and they do so by minimizing the free energy right so they have these metabolical states that have to be minimized in terms of expenditure in terms of uncertainty given that the long-term average of free energy is entropy they want to minimize their uncertainty in terms of their metabolical states well that that entails that they have a specific range of States they want to stay in like minimally just stay alive to stay alive you have to do a bunch of things as a human being but every single embodied agent would you prefer to stay alive so there are some preferred basic states that we've defined how action will be take for us to maintain it metabolical states and how that coordinate with the environment which entities that organisms and agents can be described in in those terms like in active inference tries to bring out those minimal assumptions in mathematical terms that try to express how reducing the probability of these non-anticipated states or States they are not favorable to maintaining their minimum

metabolical needs will be reduced right and that can be catch the in terms of a generative model as we've been saying throughout the best actions how this genetic model set specific times and organizes around specific parameters that wants to they can be described as trying to maintain those parameters within reasonable uh range so by minimizing free energy on average and over time this system will self-organize so we'll bring out the parameters of each internal states that occupy a limited number of possible State through set into a specific range that will then be then we'll be used as somewhat could be in the guide for this Behavior you have two kinds of Dynamics happening here right that's what I want to talk about a little bit so we have then a phenotype that exists in a specific type of States but that will have to coordinate two different kinds of Dynamics they're going to be stated here as complement complementary right you have both states you have belief updated so with Flow State you have one type of modeling which is the normal equilibrium steady state there is intrinsic is named here intrinsic information geometry and you have believed updating that can be can be treated as extrinsic information on Geometry and those are different types of modeling or the different types of thinking about those processes that's most important right so in terms of flow States we can talk about uh dynamical system and how you settle and self-organize specific parameters and in terms of updating we will be talking about by Asian mathematics and by Asian cognitive cognition and in specific ways of cashing out how an agent will change and behave over time in relation to the Dynamics that is set into place in this like State space of possible actions so while dynamical systems will explore temporary contextual dependence between environment and agent will set this specific State space that has temporal Dimension and has contextual Dimension and once this state space is sort of established we have more or less uh it's a more more or less deterministic approach to what what can possibly unfold so how changes changes in how Behavior can be cashed out in terms of non-deterministic behavior and that is defined by the different type of probabilistic thinking which is division mechanics of belief or belief updating so we have then a process of belief updating throughout weighted interactions within this range of possible States so we do have Dynamics and we do also have by Asian process unfolding together in terms of mechanisms and with that we do have mathematical statements providing a unifying perspective on formal approaches connecting behavior and Dynamics so that's what active inference should be bringing about as a novelty in relation to other forms of understanding organisms which can also can either cash things out in terms of behavior or in terms of Dynamics so trying to bring these two things together is quite interesting to understand how symbols and body semantics for instance so

that's what that's the question I will be asking right now let's talk a little bit about that I'm drawing here from a quite interesting paper from [Music] uh Maxwell Homestead and Indonesia political who is um is in is free energy principle a principle for formal semantics so what what would be formal semantics it sounds like an oxymoron right like how semantics can be formal how can you formalize it so semantics the theory of meaningful things so how can that be empty of content well it has a few principles in which you can explain what performance semantics should how a formal semantics should be understood first off we have by Asian formalism that does not necessarily have to commit to classical Notions or representations which are propositional in nature have propositional semantic Concepts so the idea is that you can use the notion of representation in a way that is not necessarily contentful so how that's going to be so under the free energy principle representations can be formal that's the argument they are bringing under the physics of uh flow specifically Dynamic ecosystem theories and information geometry you can have this mathematical statements that only represent possible uh specific ways of interacting how these correlations will affect each other in over time in this variational phase type of Cashing Out mathematical functions in this sense we can talk about representation you know how we can be understood under this formalism and what they are saying is that they are best understood as internal structures that enable the system to parse out their sensory sensitivities or the sensory streams so to organize them in specific parameters and also in terms of their own Market of blankets and in their blanket States which is interesting here which is the new the novelty of this paper is that ground in a mathematical model in these mathematical properties of the system will like avoid grounding this properties in terms of the properties of symbols and semantics in abstract or subjective Notions of meaning but impossible ranges of action so summing up what we had before was a type of computational model that had specific components that does not allow for the intelligibility the intelligibility of cognitive component which is intention which is motivation contact spring computer metaphors we have neural and cognitive activity often seen as a transfer of symbolic information and cognition then the snapshot of this transfer so it's totally dependent of that kind of symbolical transfer which is really difficult to explain how that's going to be biologically plausible and there's a lot of effort trying to do that so if you're interested in this little there's a very vast wealth of literature but they have to answer to this problem so cloud in my name brain needs to produce an impromptu response to inputs of their external environment all the time and the idea of this formal semantics the authors is trying to explain is trying to include this component into the model so you have a deflationary model

about content so you have a formal semantics we have semantics without content interestingly and formulated as a variation of free A variation free energy functional so those structures that embody those process correspond to status states in the system name and we also have the ecological component that can be formalized in terms of these two informational geometry that we the student densities do probabilistic the answer is one from the Dynamics and one from the equation describing Behavior and I watch so how can active influence be compatible to biosomatics can it be actually compatible or is it even necessary active inference is a very general framework and then it does not really give us the tools to understand how encoding can happen in terms of symbols like how they can be broadcasted into Collective action because prediction alone does not explain how do they come about so in this sense uh we do have meaningful correlations that refer to the Dynamics of one or two or more systems in terms of informational geometry that needs to be cashed out with predictive value so how is this we can talk about that in terms of shared attention that's when one form of these kind of model chip cache designs so we have active inference capturing relevance statistical relations between environment and in aging and why we why I'm trying to think about here is how biosemiotics offer tools for this informational sensitivity which is an in statistical relations to catch symbols in terms of use an interpretation within communities of language users oh well and that can only happen in the sense via directed action so the problem now will turn me to how is it possible to specify the factors that are sufficient to constrain action and directed into specific genetic rest interaction about is European forestry I'm picking this example from a very important like a speech from saying this I really recommend kind of thing tells the story about how coordinated action how State coordinates specific Collective actions but that's not the point that I want to talk here so here let's unfold that a little bit in terms of the design of scientific forestry and think about scientific forestry as a practice and a practice that has an organizing principle that we will like um well that will be the main important uh articulating story The can connect together how agriculture layouts plantations in specific ways and how Collective Collective firms will make specific choices in terms of this organization and how Villages can structure themselves around one specific notion or one specific or organizational principle and how everything can be calculated to make once one type practice which is agriculture is a very general more legible and more manipulable for different kinds of communities so in that we will both manipulating a very concrete aspects of the land the terrain the products the workforce so it has a very encompassing effect over time and in a very concrete um in a very concrete sense and interestingly when you talk about European forestry

and how European forestry was modeling the terrain and how is Cashing Out value of these terrains they were not trying to do that to successfully depict the full range of diversity of the actual social activity and also the actual activity of Agriculture or the richness of the vegetation or they are not even trying they didn't intend to do that that was not the point of this kind of modeling turned into a practice right those models generally represent specific slices of nature that were interesting for one specific official Observer and that point is really important because that we came how all these communities and how the terrain and how the forest we won't fold over time within this specific perspective so I bring here a quote that I really really like because it tries to flesh this out in simple terms and very useful ones so thinking that exaggerating only slightly one might say that the interest in forests was resolved true when fiscal uh specific single number which was the revenue that you from the timber that was extracted annually well that's okay that's understandable like that's intelligible and that happens with you every day in terms of how we understand our economy but what is interesting about this quote is that the best way to appreciate how heroic was this constriction of vision is to notice what Falls outside the field of vision so what is missing from that is not the the organizational principle of Timber being valuable but everything that was not there or was being excluded treason bushes and Planters implants that hold little to non-potential for that purpose so missing out of that is Florida diversity foliage fruits Etc twigs and branches missing out of that are insects and small animals depending on this diversity all this ecological organization falls out of the organizational principle and that has very complete consequences that we still suffer and we still live we live in the height of this climate crisis that also starts from this kind of thinking so it has really very material consequences for us and had for them at the time if you read more about it you know that a team whereby like ex by dreaming and domesticating nature so much like decreased in quality and then eventually decreased in value which prompted them to reach to try to reorganize that kind of story and what do I want to call attention with the example is that also it brings some specific ways of talking about it it brings vocabulary or that only exists in terms of this type of practice specifically so this vocabulary constrains nature in a way by pruning what can what cannot be used so it generates more than illegible nature but a way of making nature intelligible and intelligible in a very specific certain way and we make that happen in terms of symbolizing things in specific terms but the symbols they don't exist in themselves they exist only in organizing this social practice around them so when you replace nature with a term called natural resources we are doing exactly that it's specifying from a generalized notion of nature like flora and fauna becomes something

valuable becomes a commodity but not everything right it trims the practice it separates competing species from value guns so plants are symbolized as crops and what is or the competing species are symbolized the weeds they are not necessarily crops or weeds or anything like that in nature but area that for the way we talk like that because we agriculture in a certain way that also have that that's the same for insects or small animals that consume these interesting crops the material effect of that and how that only gains Traction in the practice over time and how you have specific intention that has unintended consequences uh are necessarily here situated contextualized and embodied in terms of trimming specific types of attention and this attention will only shift to reach to try to repopulate insect population for instance when you realize that this practice like brought this specific effect in a certain way that affects the value of our understanding of nature and that generates expect exactly a type of social coherence and social coordination and how constraints create these coherence right now let's talk back again like cement about formal semantics while while these formal semantics and active influence sort of embeds one type of mathematical formalism symbols can only be grounded in a specific concrete practices or in action so while active inference can be used as a tool to explain social understanding it it does do that in terms of generalized synchronization okay in a model of cognition that can be inactive and dynamic but is it still very general in terms of biosemantics we can think of it as a framework that might be compact compatible with active imprints in terms of explaining the materiality of the environment in a non-reductive in a non-reductionist way that tries to retain some naturalistic credentials so how that conventions are affecting and effecting the invited agent in terms of forming those singles and cashing out those epistemic values well social interaction in this sense is not reduced to prediction because in the sense the prediction will not be a primitive but it would print but it would be a result of a specific participatory construction of the social scene over a specific organizational principles and actions through the general through this synchronization of this generative models or the synchronization of those agents that can be described in terms of those generative models gain predictive value and when you model that then you have this type of understand the value of prediction in those senses that allows that allows us to say that social actors in distance can behave as if they have a generative model that synchronizes with that symbolic State of Affairs allowing them to understand each other's behaviors or to develop approximate sales informational sensitive or responsiveness to the same symbols and goals they are approximate because they're obviously not going to be the same but they have a they have the same approximate orientation so in this sense we don't have ever a view from

nowhere we don't have ever this view that is only cast out in terms of specific types of minds but you also have types of Mind shaped in terms of this very concrete organizations that happens in the community like for humans at least it happens in community of language and users so the relationship with a symbolic cues helps to select attention at that specific time as well pruning what is what should be ignored is socially constructed environments so within a specific community of language users some types of actions will be desirable and will be deemed as true in that specific type of community and can be only evaluated under those terms so you can provide reasons for actions in that specific in that specific context context and you can talk about how rational that action is in terms of the conditions set by those communities in and that brings us to the idea that human cognition in this sense is never decoupled from this cultural Norms that like pushes and orients them over time so you know scale of behavior and that's the interesting thing and that's why active inference can be an interesting tool to model this type of human communities and this type of behavior because it can come from the very Dynamics the very basic of Dynamics to specific types of perception and scale up and more so when you have the problem of symbols like in these two decoupled community of users and symbols and embodied agents and things that do not combine we can try to model our agents in a way that we don't have this problem anymore because you're trying to include that very that very component into the model so in the sense we have active inference used as tools to explain social understanding as generalized synchronization and in understanding others in that sense because of this generalized synchronization model and it's assumed by those types of organizational principles we can also engage in terms of Niche construction so we have more or less deterministic way of understanding the nature of symbols because the more we interact and approximate synchronized with the environment we change the environment that might changes back so you have an open system that is always an everywhere situated and he can be modeled then with this type of predictive value that's more or less what I had in mind and that's how I was trying to think about biosemiotics and in a possible relationship with um free energy principle and the formalisms of effective inference and yeah I think this is it awesome thank you Lorena I'll just organize things a little in my room but thank you very much for this excellent lecture um just as a kind of warm up how did you come to see it this way what what brought you to this general question of semantics and how do you see that like on your own path and see what what other comparable or analogous approaches have been done in terms of a meta-analysis and proposal related to the Naturalization of semantics uh I think it's kind of hard to naturalize I think there's a lot of

effort in doing that and it might be that if you think differently about how we naturalize those values we might have better answers instead of trying to instead of trying to fit Gage and into the model we try to fit the model into the agent right I think that's the move I think that was the motivation I think the interesting the interesting part of biosemantics in that sense it tries to make less assumptions when you're we will start from a hardcore assumption or model assumption right that semantics is within only within the mind then we have a very hard time trying to explain that if you assume less about that you might bring different types of explanations of course but they might be very they'd be more interesting that I was trying to like see how that actually how that looks like when you you bring these two things together in in the terms of active inference and free energy principle if that works if that is possible I don't know if that answers your question so wondering about that fundamental question and then coming across active inference or how elsewhere have people brought in the more formal and Technical into this question what do you mean by Mar like formal and can you repeat that please yeah how does active inference help us or have continuity or difference with previous ways that people have connected the technical part from the um yeah what you described as the formal semantics embeds the mathematical formalism and then the symbols are grounded in specific action so has that been played out in other technical Frameworks well we do have technical Frameworks that try to organize that in developing activism right and auto poetic and activism but then we just have in these terms the notion of symbols uh sometimes get get Loosely described and I think these guys are not really interested in that I think they're trying to describe something else for example or you can have some specific types of computational that like if you if you really buy into the mind the computational brain and Temptation mind metaphor then you have a hard time also to explain how semantics are built in into that computation right I don't think anyone has a very strong answer to that you have to really buy into the idea of information processing and that's something I'm really that that's what brought me a little bit to that so if you buy into the metaphor of information processing then you need to have semantics within the brain if you try to lose a little bit this notion if you try to understand process and information in different terms what happens I think that was more or less the question that I was trying to ask an effective inference and bio semiotic skin because they have less assumptions in terms of what happens inside the processing information content in the mind if that can if can if that can bring interesting like answers or if you can use as a model to organize Behavior and there's another point that also brought me uh to this type of thinking was trying to understand how active active inference can include his historicity of the

agent right because I think that something comes across debated in an active influence how you bring novelty into the state space like once you define that state space the state space is more or less deterministic because everything will happen within that specific range so how does another trick comes within this type of uh modeling so how can you explain that if it does I don't know it's more like it's more like an exploratory question that I was trying to make and see how that pans out I don't I actually also don't have an answer for that but if you take seriously some more inactive claims that you need history of interactions to to understand how agents behave you cannot understand agents without that so how does active inference sensor desk these questions specifically right because once yeah that system I think that is still kind of debated I'm not sure if there's new developments in this specific topic but I remember I think two years ago was like this specific papers from the Paolo I think the 4K laying down a path is specifically focusing on that in the formalisms and Etc so I don't know if these two things can bring a little bit of more to the conversation maybe not this great I'll just ask some some more general questions you're adding a lot with these thoughts so you said the social interaction in this account is not reduced to prediction so how would you unpack that and how would you relate this I don't know exactly what term of art would go here but this kind of social epistemology that active inference provides Us in contrast to say predictive processing so how is it that that we would or wouldn't reduce something to prediction and then what is it that active inference brings into that discussion yeah oh I think I have two points in that like active interest in that sense that has like a type of modeling that tries to include all the time and everywhere the embodied agent like if if you take seriously the idea of the model the genetic model is always accurately modeling that specific agent and because you can it cannot be true false that the model is your current state like is like that like even if you are not feeling really well you just know that you're not feeling well in relation to something else and that that story is still very accurate to your stage condition right now right so the prediction you get instruction in socially at least in that sense not in the specific detail of like predictive predictive processing you think because I think that's more complicated but socially in social cognition at least we can talk about this predictive value I think is a more interesting way to do that because the mother the the example that I use in terms of Cashing Out symbols we are we're using some something that was meant in to make something intelligible that have very bad consequences that were not even close to be predicted so in that sense and how these consequences happen don't prediction cannot be there everywhere and all the time the prediction is this has an orientation as well right that's gonna unfold only in terms of the

orientation but that's not all-encompassing so well what does that say about the prediction that's why I was trying to think of it as not being a primitive but something that comes along with this Dynamics but yeah well this is a very deep and prescient point that because the causal structure of the world changes historiosity agents change they learn and they attend and then the niche changes with stigma G and Niche modification so because things really do change Through Time then just training a data set at one time point and then using that same generative model and carrying it forward it's like picking a speed in a direction when you're on a Winding Road and just saying all right now let's turn on cruise control right now and just keep going in this direction it's working right now so then to connect that to the legibility of the forests as the causal model changed and externalities were not taken into account and so on all these different features you brought in so now is active inference and cognitive modeling cognitive Technologies increasing our legibility of perception cognition and action through surveillance technology and other mechanisms such that there could be slash we expect there to be also hidden Dynamics and externalities that we are also failing to see when we make some aspects of cognitive systems legible in the way that research in the neurosciences is happening towards yeah I focus is everything right it would be non-informative in terms of what you do when you do that you have one interested thing you're interested in something right you have one thing in mind when you did you're trying to flesh out in terms of cognitive modeling how that so how that applies to these more all-encompassing situations I think that's like that model is how I'm trying to ask this question so you can model them in dislike sliced sense and that's that's what we that's what we do as scientists and that's how we learn about but that only way of seeing that that very specific model does not all encompassing for Owen so it's like so how can you how can you use that in comparison to other situations so how can you integrate this this type of thing I don't know I really I I don't have a question for an answer for that I don't know if we do have an answer for this this type of question currently but I think it's a very valid question to ask because because also in the terms of sometimes applying that socially that sort of revised in a way and then it becomes someone organizing principle that becomes somewhat reified so keep an eye for this kind of metaphorical use like assuming a little bit that all of this talk can be metaphorical and it can be interesting and useful to catch that out catch them out in these terms because they don't have other ways to talk about them how we don't reify them and how they are in you know because well I think I'm I'm getting I'm having a hard time to get there but I think I think I'm making the point of people criticizing the cause of Primacy so you when you reduce this complex systems to one single cause

cashing out everything that organizes the principle this this these complex systems and when Primal cause then you are losing the complexity however if you don't do that how can you slice it out you understand the system so I think that's the two tensions here that I'm kind of trying to to to propose as a question in term instead of an answer I don't know if that makes sense but I think yeah I hope people continue that discussion I'm gonna build on that just to kind of maybe circle around that because it's really important so in the forestry case the the realist perspective again using retro casting the way we or contemporary discussions some of them in the active inference and free energy principal space are the realist says that region of land is just economic potential it is a natural resource that becomes a kind of handle the instrumentalist perspective takes a step back and says it's as if that piece of land is a natural resource so yeah the territory is Indescribable and there's still a space for mystery but then when the proxies come out because policies are being selected the as if in some ways carries similar consequences to the actual because they and they may totally agree and the realists and the and the instrumentalists might be like yeah it is an economic potential or represents or it could be treated as such or it could be engaged with as such like that cognitive diversity can align on the social policy that then connects to these broader social questions and histories and so it's kind of interesting like what work does this instrumentalism do like oh it's just a what work does it do and what doesn't it do and the ways that that science and technical engineering fields are and aren't deployed alone or together with other considerations yeah that's very interesting like in the sense if you're realist or instrumentalist about that like if you think like models are not either true or false they're just models right they might have views and you might they might be capturing real properties but the problem is in this conversation even when you push them like have realist and have an instrument let's just try to like loosen up a little bit these assumptions they still come they still end up refine those principles and taking them to be more or less true or false right well that's kind of plot twist it's like you're appealing to what surprise minimization as unifying the diversity responses that the realistic instrumentalists do so then does that make surprise or cognitive inertia like the ER principle where the maze just ends and it explains itself or how does that play in a pluralistic scientific setting yeah that's that that's you you you you might you can have a third answer to that like right I have two well I think the answer that I'm trying to give is more eliminativities in the sense that eliminates the cause of problem Primacy of them of these two modes of answering when you think I'm thinking I have in mind now like actually just to try to make sense of that when you when you think about biological information you can think about biological

informations being rewarded as being metaphorical or instrumental or you have these guys like Susan Oyama or Richard Levins in Detroit to bring some they're against this cause of Primacy they try to advocate for systems that do have that but they have like a lot of more going on so that cannot be the only answer because those systems is not everywhere scientifically cash they don't have to be everywhere that so for example going back to to the forestry forestry so the realistic is going to say that territory is valued properly and the the in that sense the instrumentalism look as if and it can be he is as if but it's not necessarily value property it doesn't matter you can say like well this terrain is theme from one perspective as value property but this also has other uses around them so you had to be more inclusive of this diversity because this the only organization principle around that specific terrain is not only to generate value or if it does this value doesn't have one single output it can have more than one output in terms of very specific terms you can catch it out very maturely this terrain that has to provide monoculture needs to also has to provide also value for the diversity of the fauna and the diversity the diversity of the Flora around it and that is not included in a model because that's not what a model is done for the model has a specific use and it's fine however that does not reduce the terrain to that and well when you're talking about cognition that might be more complicated but I think it's easy it's an easy way to talk about this causal Primacy when you talk about uh forestry because we are very much now experiencing this climate crisis and we see these things happening right now and how difficult because this this type of thinking is built over time in so well fixed in the way we see things and we are understanding how hard is to change that it's not it's a it's a paradigm shift that needs to happen for that to change is not necessarily the model that is right or wrong it's just how you understand and apply them so there are pragmatics into that it's not only an instrumental but it has something that the instrumentalists take does not capture and I don't know how to catch it out in a one single word but that's what biosemantics was offering biosomatics was offering the idea of practice in pragmatics I don't know if it's the whole story but it's another Point into that story a lot of great points all all connect that no of pragmatism to a little bit of the active inference formalism so when it comes to policy selection Behavior selection some but not all cognitive entities can act as if they're calculating expected free energy so these are the active the truly active entities whether they're simple or whether they're sophisticated they do planning expected free energy is how those agents are being modeled as if they're making decisions and it consists of the pragmatic and the epistemic value that's how these different policies are evaluated and so I see two angles or windows that that supports at least for my non social

sciences background a new kind of meeting in the middle there so first is the expansion of pragmatic value under the principle that surprise minimization or bounding is the imperative right there in the pragmatic value it doesn't have to be just body temperature or financial income we can expect and prefer ourselves to be learning new things or meeting new people and so we can load into the pragmatism different goals that under a narrower definition of utility or pragma have not traditionally been included so I think that first way is it speaks to a more inclusive concept of utility you might say I'm also de-risking on forest fires alongside the Timber and already you've had you have now plural pragma so I think that's the kind of crossover and then the deeper and second and not saying there's only two ways that that the conversation continues we have epistemic utility in terms of epistemic value and so the balance and the weighting between pragmatic value and epistemic or learning oriented activity overall helps control a dynamic balance with just like how tight is the grip and the preference overall and maybe we have uh some preference for Timber to be exported from a region but we really don't know we don't have long-term modeling we don't have deep causal analysis and so we're going to put five percent on the pragmatic extraction and other pragmatic features and leave the rest of the policy in an exploratory mode and come to a blend that's not just on the the Spectrum from explore to exploit which is how this has been approached in like cognitive Sciences previously it's like this is the third thing where policies that are on the frontier of exploring and exploiting rise to the top through their contributions to pragmatic and epistemic value and we made the model so that isn't to say that that is what's happening on the ground or maybe that's just the instrumentalist perspective yeah yeah but if it's an instrument an instrumentless perspective it might be a good one right in the sense the idea of inclusion is kind of the interesting one right because if you really find a destination of maximization because you can use that in certain ways to understand what happens within when you're modeling something what is the limit of that maximization so that's you can model agents as if maximizing utility for instance but maximizing utility is something that needs to be balanced out with utility of everyone around them right and that's not that's not trivial how that happens right and even if it does and even if it were travel it's not something that Maps out completely identically with all every single agent they don't that's not that's not going to be there's going to be diversity there so how do you make that notion more inclusive and less verified in that sense because then if the more you refine the more you single it out into around only one General type of understanding so when you think them more individually then you kind of lose you lose that strength of this one single organized like the one single way of Cashing Out

fertility that's not the principle like sorry so I think that's like speaks a lot with lightning the type of instrumental use and the type of pragmatism and how the time the types of application of this practices right remember in in the sense of the scientific forestry it's also a practice it's not something that is true in itself it's true in terms of practice right in terms of applications in terms of building communities around them so without this use they have no grounding yeah one really strong and Salient point that I got from the lecture was about generalized synchrony and it's almost funny that we have to Advocate or delineate generalized synchrony because the alternative the absence of generalized synchrony is just like kind of like an ideal gas it's like there wouldn't be regularities in society we wouldn't be clustered in space or in time we wouldn't have information and communication and we wouldn't have what differentiates a society or a group or civilization or a colony or whatever from literally anything else so generalized synchrony is almost like all of the space between nothing and anything happening and then it supports all these different scenarios we could have this kind of mind in this kind of brain in this kind of environment this kind of Niche and so we can talk about the ant colony foraging algorithms that work in the desert and there might be mutual non-exhausted open-ended change of algorithms and what works in a different setting and with humans we have the sophistication and the history and the scope and the scale and all of these metas and weaving with is an ought and our own forward going Direction which is really the moment where it goes from being a forensic sociology to the real time sense making an action selection based upon our cognitive models in the broader sense which includes humans and non-human agents but it's like that's when it goes from being more than just an understanding when that understanding translates to an action policy and they do then it matters as much as anything does yeah well one of the critics criticism of the active inference I think it's like it's too General to single out specific constraints that Orient action that's what I saw like thinking about how symbols are brought about in community of new language users assuming that the symbols are not they they are only meaningful throughout this histories and practices can help us to apply this type of General story right yeah navigating specific uh well epistemic communities and I think like that's the work that my whole is going to bring along in terms of epistemic communities and then you single out in them so you have these communities of language users in which specific practices create a specific orientations yeah one thought on that asking like what is uh active inference or linear regression or some other method again this is who knows is this just deep esoteric instrumentalism but asking what does it have to say about this symbol what is what does active inference have to say about the letter E in English

can it be as simple as to say that that's a category error and that you could make a toy example or a pedagogical example where symbol one and symbol two oscillate in a multi-agent simulation or you could make a generative model for an alphabet and leave it abstract but you're never going to connect to the historicity and the real enactment of the English letter e unless you're confronting the my new particulars and the actual embodiment of that letter and then there's different core screenings different iterated modeling decisions different questions about how do you include and participate in a different stages of the scientific process but it's like you're in the specifics at that point active inference doesn't have an opinion of the letter e because we wouldn't want it to it would be too opinionated about the wrong things it'd be making and bringing axioms into play that would be extremely constraining and so it remains General as a framework and the free energy principle even more slash differently so but when we get into the actual that's where we're connecting the real systems of interest but how how can we make claims and Advance them socially about systems of Interest when we're not even it's like comparing bridge building as a concept with this bridge right here it's a little bit like a category disconnect well it depends on also how you single out some things they are not they're just you just don't do that because they're just not interesting enough right there's nothing like that comes out of it in terms of this bridge right here in a single how did you put it again like your category of bridges in this bridge right here right when you do that you can learn something about a category in terms of this single one right here but he does not it's not exhaustive that's it that's the thing one point it's not exhaustive and it's not right or wrong it might be better than other I think they did like you can bring some com you can make some comparisons in terms of like predictive value you can have like better or worse ways of doing it and then it gains more utility okay connecting this to also um this this non-exhaustive nature and where the power of a generalizable framework is let's talk about the category of Golden Gate bridges in San Francisco so there's only one actual Bridge there but we're generalizing it and so somebody makes a state space and says well I'm exploring if there was more than two end to end so they're exploring the state space of topology is a bridge and someone else is exploring the state space of color and somebody else is exploring the state space of traffic Dynamics and someone else is combining two of them or combining all three of them and then that's the conversation is which map of this category of State spaces category of adjacent possibles as we chose to model it and if you don't include traffic flow don't be surprised that it's not going to take that kind of data or make that kind of prediction and some variables matter and some matter in combination and some don't but when we model we kind of build

these regions or adjacencies to the actual which is where the historicity of the past leaves us and then we have an adjacent possible as modelers and sense makers including absurd counterfactuals and state spaces and then the state spaces are mobilized yeah in a way but if you talk about like specific type of engineering yes in that sense they are mobilized and they are not true or false you can use false models that is to function that's not like you can have specific uses of things that you know they're not really true like I think the tolerate totally caloric type of modeling the uh social uh the solar system is to use that to navigate you know in the desert even though it's not really accurate but if you want to go from A to B it helps you to do that that's a different thing right like it doesn't it's not true or false but when you think about agents in terms of like and if bringing back to the idea of uh symbols being only cashed out in terms of community fuses that have histories and they they are made useful they have made meaningful in terms of their then you don't have ever this disconnect right you have a comment you can think about this cognition as always connected so there's a specific types of practicing you can even single out some moments of it and try to see what you learn from this kind of slides of time but when you think about the whole story of the cognitive system you can if you're gonna really think of it in terms of is valuable you cannot disconnect from the state let's say solid state space of it s practice awesome well so do you have any closing thoughts or what are you looking forward to with the discussion section the rest of the course just by way of wrapping up and letting people know any other thoughts yeah we go to explore these these ideas that we've been actually bringing along here right how we can go with that like instrumentally realistically or how can we understand this General synchronization of the model in terms of instantiations and how formative they are is that really interesting or how candies things be actually refined modeling and if that happens does that have any specific consequences actually or not right maybe not awesome well very ambitious and excellent lecturer so thank you very much Lorena and yeah just yeah just keep in mind it's just I was just like trying to ask these questions I don't also have those answers like it's I I found very little about active inference and semiotics like produced I found like a few papers so that's very exploratory actually so yeah so you're invited to think of these things and explore a little bit then further if it's possible yeah oh epic all right thank you learn to see you in a couple of weeks farewell see you bye